Skin Cancer Detection

ISIC 2024: 3D-TBP Crops & Metadata Fusion

Project: Deep Learning & Advanced Machine Learning

Phase I Report

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Abstract

Population-scale skin cancer screening demands AI systems that are accurate, calibrated, and clinically reliable. We develop and evaluate four approaches to binary malignancy detection on the ISIC 2024 SLICE-3D dataset: **(1)** a metadata-only LightGBM baseline, **(2)** classical computer vision feature extraction, **(3)** an EfficientNet-B4 deep learning classifier with Focal Loss, and **(4)** a hybrid fusion model combining visual and clinical information with temperature-scaled probability calibration. All models are evaluated using partial AUC at TPR $\geq$ 0.80, reflecting the clinical requirement to maintain high sensitivity. **Our best model achieves pAUC = [TBD] on held-out validation**, compared to [TBD] for the metadata-only baseline.

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1 Introduction

Melanoma and other skin cancers represent a major global health burden, with approximately 57,000 deaths annually worldwide. Early detection dramatically improves outcomes: the five-year survival rate for localized melanoma exceeds 98%, falling below 30% for metastatic disease [8]. Despite its importance, dermatologist availability is highly unequal globally, making AI-assisted population-scale screening a compelling research direction.

This project addresses the 2024 ISIC (International Skin Imaging Collaboration) challenge [6], which reflects the real-world complexity of screening via 3D total body photography (3D-TBP). Unlike classical dermoscopy datasets, 3D-TBP introduces variable lighting conditions, diverse imaging angles, and lesions embedded in full body context rather than isolated close-ups.

1.1 Problem Definition

Given a crop of a skin lesion from a 3D-TBP scan and a vector of patient metadata (age, sex, anatomical site, and TBP-computed morphological features), predict the binary malignancy label $y \in \{0, 1\}$.

The dataset exhibits severe class imbalance: approximately 3.5% of lesions are malignant. Standard accuracy is therefore a misleading metric — a trivial classifier predicting all benign achieves 96.5% accuracy while missing every cancer. The clinically appropriate metric is the *partial AUC* (pAUC) restricted to the region where true positive rate (sensitivity) ≥ 0.80, formalized in Section 4.1.

1.2 Contributions

This work provides:

1. A systematic comparison of four methodological pathways with identical evaluation protocols.
2. Domain-informed feature engineering motivated by the ABCD rule and clinical risk factors.
3. Patient-aware cross-validation (`GroupKFold` by patient ID) that prevents data leakage from multi-lesion patients.
4. Post-hoc probability calibration via temperature scaling for clinical reliability of model outputs.

2 Related Work

2.1 Deep Learning for Dermatology

Esteva et al. [3] demonstrated that a CNN trained on 129,450 clinical images matched board-certified dermatologist performance on binary classification of keratinocyte carcinoma vs. benign seborrheic keratosis and melanoma vs. benign nevi. This landmark work established that ImageNet-pretrained features transfer effectively to dermoscopy, forming the backbone of our deep learning approach.

A critical gap in Esteva et al. [3] is the absence of patient metadata. Clinical diagnosis integrates visual and contextual signals — age, anatomical site, and lesion history are routinely used by dermatologists. Our hybrid approach addresses this gap.

2.2 ISIC Challenge Series

Codella et al. [2] introduced a structured multi-task challenge framework for skin lesion analysis, providing standardized training and evaluation protocols that this project builds upon. The 2018 challenge established the pAUC metric as the standard for clinical screening evaluation.

2.3 Focal Loss for Imbalanced Classification

Standard binary cross-entropy loss is dominated by easy negative examples when class imbalance is severe. Lin et al. [7] proposed Focal Loss in the context of dense object detection:

$$\mathrm{FL}(p_t) = -\alpha_t \, (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

where p_t is the model probability for the correct class, $\gamma \geq 0$ is the focusing parameter (concentrating loss on hard examples), and α_t balances positive/negative classes. We use $\gamma = 2.0$ and $\alpha = 0.25$ following the original recommendation [7].

2.4 Probability Calibration

Guo et al. [4] showed that modern neural networks are systematically overconfident: a network predicting $p = 0.9$ may only be correct 70% of the time. Temperature scaling is a simple post-hoc calibration method that divides logits by a scalar temperature T:

$$\hat{q}_i = \sigma\left(\frac{z_i}{T}\right) \tag{2}$$

where T is fitted by minimizing negative log-likelihood (NLL) on a held-out validation set. For clinical deployment, calibrated probabilities are essential — physicians must be able to trust the numerical risk scores provided by the model.

2.5 Gap Analysis

Table 1 summarizes the methodological gaps that this project addresses relative to prior work.

Table 1. Gap analysis: prior work versus this project. Ticks indicate that the property is satisfied.

Property	Esteva 2017	ISIC 2018	This Work
Image-based deep learning	✓	✓	✓
Metadata integration			✓
Calibrated outputs			✓
Patient-aware CV			✓
3D-TBP data			✓
pAUC evaluation metric		✓	✓

3 Dataset & Exploratory Data Analysis

3.1 Dataset Description

The ISIC 2024 SLICE-3D dataset [6] contains approximately 400,000 lesion crops extracted from 3D total body photography. Each sample consists of:

- **Image:** JPEG crop of the lesion region (224×224 px minimum).
- **Metadata:** ≈40 features including demographic information (age, sex), anatomical site, and TBP-computed morphological indices (color statistics, shape metrics, symmetry scores).
- **Label:** Binary target ($y = 1$ malignant, $y = 0$ benign).

3.2 Class Imbalance

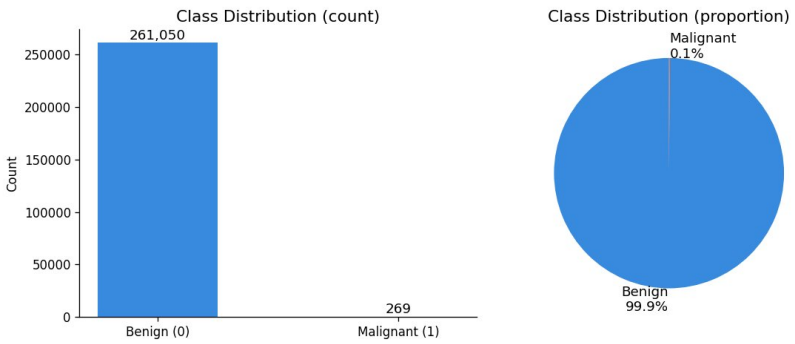

Figure 1. Target class distribution. The severe imbalance (≈3.5% positive rate, ≈27:1 benign-to-malignant ratio) motivates the use of Focal Loss and pAUC as the primary metric.

As shown in Figure 1, the dataset exhibits a $\approx 27:1$ benign-to-malignant ratio.

Modeling Decision

This directly motivates our use of Focal Loss (Equation 1), the `is_unbalance=True` setting in LightGBM, and validates the choice of pAUC over accuracy as the primary metric.

3.3 Patient-Level Structure

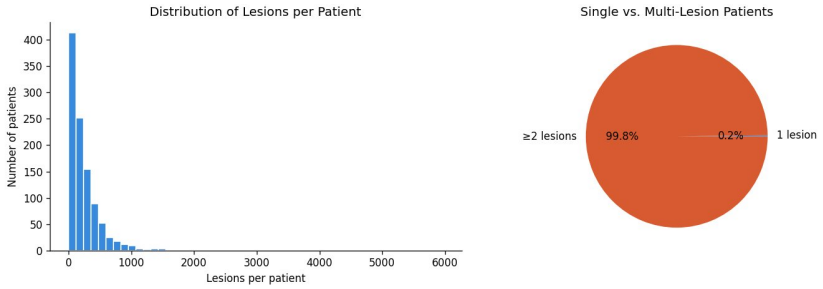

Figure 2. Distribution of lesions per patient. Many patients contribute multiple lesions, creating a data leakage risk in naive random splits.

A critical finding (Figure 2) is that many patients contribute multiple lesions. A naive random train/test split would allow the model to implicitly learn patient-specific characteristics (e.g., skin tone, imaging setup) that are correlated with malignancy but would not generalize to new patients.

Modeling Decision

All cross-validation uses `GroupKFold` with `groups=patient_id`, ensuring all lesions from a single patient appear in the same fold.

3.4 Feature Distribution Analysis

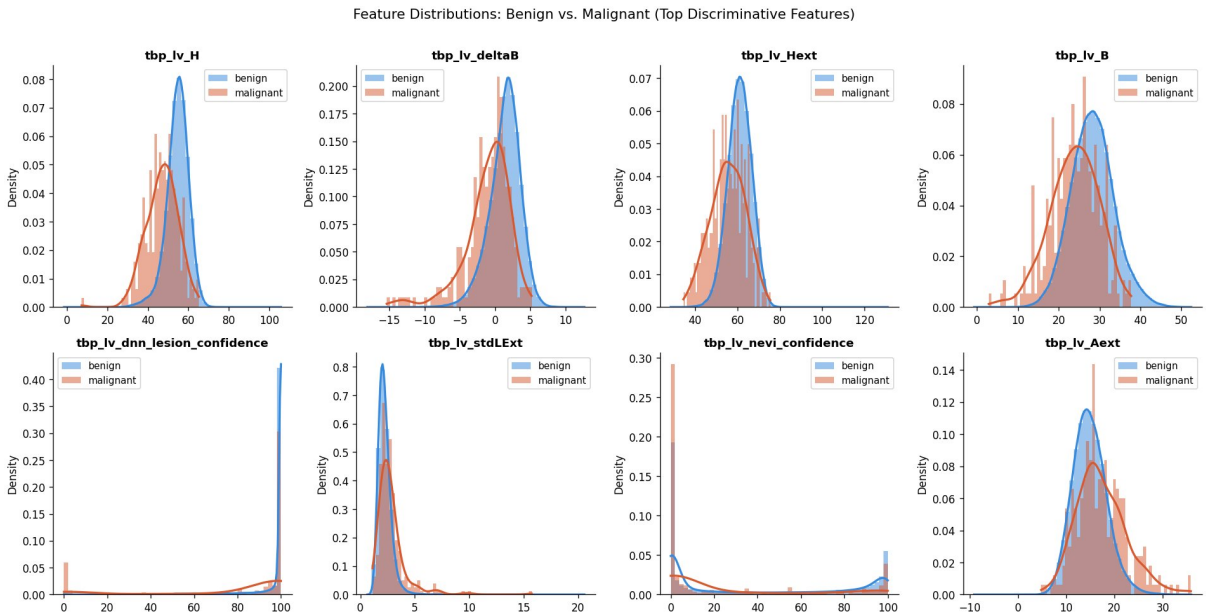

Figure 3. KDE plots of the top discriminative metadata features, stratified by target class. Distribution differences between malignant and benign classes motivate domain-informed feature engineering choices.

We applied the Mann-Whitney U test (non-parametric, robust to non-normality) to assess the discriminative power of each numeric feature (Figure 3). Features showing heavy right-skew received a $\log(1+x)$ transform to improve LightGBM split quality.

3.5 Image Quality Analysis

We audited 100 sampled images per class using Laplacian variance (sharpness proxy) and mean HSV-V channel brightness. The distribution of brightness was notably wider for 3D-TBP images than for classical dermoscopy, confirming the need for aggressive color augmentation.

Modeling Decision

Training augmentations include `RandomBrightnessContrast`, `HueSaturationValue`, and CLAHE to simulate 3D-TBP lighting variation.

4 Methods

4.1 Evaluation Metric

The primary evaluation metric is the partial AUC (pAUC) normalized to $[0,1]$, restricted to the operating region where TPR ≥ 0.80:

$$\text{pAUC} = \frac{1}{1 - \text{min_TPR}} \int_{\text{min_TPR}}^{1} \text{AUC}(\text{TPR})\, d(\text{TPR}) \tag{3}$$

This metric quantifies discriminative ability *only in the high-sensitivity operating region*, reflecting the clinical constraint that a screening model must not miss cancers (false negatives are more dangerous than false alarms).

4.2 Validation Strategy

All models are evaluated with 5-fold `GroupKFold` cross-validation grouped by `patient_id`. Out-of-fold (OOF) predictions are aggregated to compute the final pAUC, which is used for model comparison and hyperparameter selection.

4.3 Pathway 1: Metadata-Only Baseline (LightGBM)

4.3.1 Feature Engineering

We apply the following domain-motivated transformations:

- **Age risk groups:** Melanoma incidence increases sharply after 50. We create a 5-level ordinal risk group and a continuous risk score $s_{\text{age}} = 1 + \frac{\text{age}-50}{50}$ for age ≥ 50.
- **Anatomical site risk tier:** Palms, soles, head/neck, and oral/genital regions carry higher malignancy rates. A 3-level risk tier is encoded per site.
- **Age $\times$ site interaction:** Multiplicative interaction captures compounded risk in older patients at high-risk sites.
- **Color variability index:** Standard deviation across TBP color features — higher variability indicates color asymmetry (ABCD rule).
- **Circularity:** $C = \frac{4\pi A}{P^2}$ from TBP area/perimeter features. Low circularity indicates irregular border (ABCD rule B).

4.3.2 Model & Hyperparameter Tuning

We train a LightGBM classifier [5] with:

- `is_unbalance=True` for automatic class-weight correction.
- 5-fold GroupKFold with early stopping (100 rounds patience).
- Optuna [1] hyperparameter search (50 trials) maximizing OOF pAUC.

4.4 Pathway 2: Advanced ML (Classical CV Features)

We extract the following handcrafted image features:

- **HOG** (Histogram of Oriented Gradients): shape/texture descriptor.
- **LBP** (Local Binary Patterns): local texture encoding.
- **Color moments:** RGB and HSV mean, standard deviation, and skewness per channel.
- **GLCM** (Grey-Level Co-occurrence Matrix): contrast, correlation, and entropy descriptors.

These are concatenated with metadata features and fed to SVM and Random Forest classifiers.

4.5 Pathway 3: Deep Learning (EfficientNet-B4)

We fine-tune EfficientNet-B4 [9] with ImageNet pretrained weights using:

- Focal Loss ($\alpha = 0.25$, $\gamma = 2.0$) for class imbalance.
- Albumentations augmentation pipeline motivated by EDA Section 3.
- AdamW optimizer with cosine annealing and 2-epoch linear warmup.
- Mixed precision training (FP16) for efficiency.

Bias-Variance analysis. EfficientNet-B4 has 19M parameters — significantly overparameterized relative to our dataset. We address high-variance risk with: (1) ImageNet pretraining (strong prior), (2) Dropout ($p = 0.3$), (3) strong augmentation, and (4) early stopping on validation pAUC.

4.6 Pathway 4: Hybrid Fusion with Calibration

The fusion model concatenates embeddings from both branches:

$$\hat{y} = \sigma\left(\mathrm{MLP}_{\mathrm{fusion}}\left([\mathbf{f}_{\mathrm{img}}; \mathbf{f}_{\mathrm{meta}}]\right) \Big/ T\right) \tag{4}$$

where $\mathbf{f}_{\mathrm{img}} \in \mathbb{R}^{1792}$ is the EfficientNet-B4 embedding, $\mathbf{f}_{\mathrm{meta}} \in \mathbb{R}^{64}$ is the metadata MLP embedding, and T is the temperature scaling parameter fitted on validation data.

5 Results

5.1 Model Comparison

Table 2. Model performance on 5-fold GroupKFold out-of-fold (OOF) validation. Primary metric: pAUC @ TPR $\geq$ 0.80. Higher is better. [TBD] entries will be filled after training.

Model	pAUC ↑	AUC	Sensitivity	Specificity	PPV	F1
Majority-class baseline	0.500	0.500	1.000	0.000	0.035	0.068
Metadata LightGBM	TBD	TBD	TBD	TBD	TBD	TBD
CV Features + SVM	TBD	TBD	TBD	TBD	TBD	TBD
EfficientNet-B4	TBD	TBD	TBD	TBD	TBD	TBD
Hybrid Fusion	TBD	TBD	TBD	TBD	TBD	TBD
Hybrid + Calibration	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

5.2 SHAP Feature Importance

SHAP feature importance analysis for the LightGBM baseline will be presented following model training. Features will be ranked by mean absolute SHAP value, with direction of effect on malignancy probability indicated per feature.

5.3 Calibration Analysis

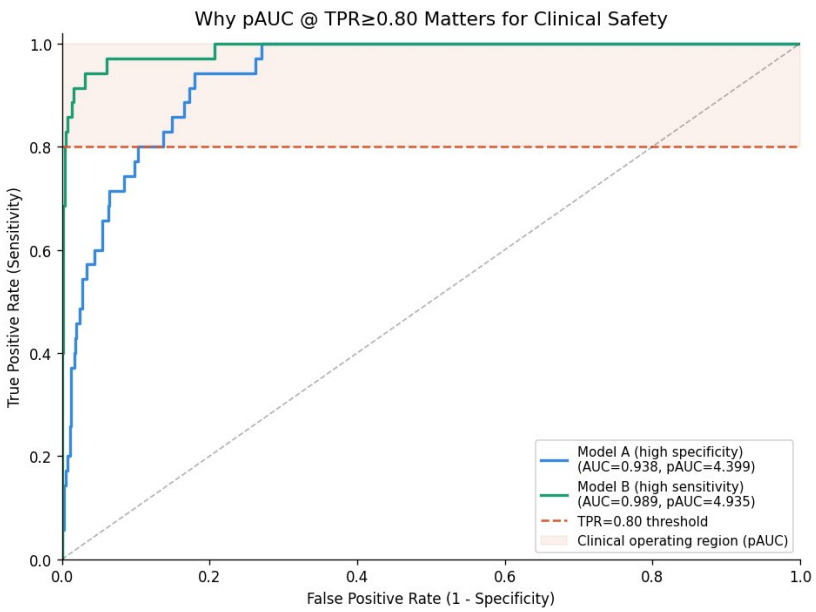

Figure 4. ROC curves illustrating the clinical operating region (shaded) corresponding to pAUC @ TPR $\geq$ 0.80. A model performing well only in the low-sensitivity region is clinically unacceptable.

6 Discussion

6.1 Failure Analysis

Common failure modes in skin lesion classification include:

- **Seborrheic keratoses:** Benign lesions that visually resemble melanoma, causing elevated false positive rates.
- **Amelanotic melanoma:** Lack of pigmentation makes these malignancies appear benign even to clinical experts.
- **Low-quality images:** Blurry or poorly lit crops from 3D-TBP degrade classification confidence.

A detailed quantitative failure analysis will be completed following model training.

6.2 Theoretical Considerations

IID assumption. The i.i.d. assumption is violated in this dataset — lesions from the same patient share correlated features (skin type, imaging setup, genetic factors). GroupKFold mitigates leakage at training time, but deployment on new patient populations may still exhibit distribution shift.

Bias-Variance tradeoff. The metadata-only LightGBM model operates in the high-bias regime (limited information from 40 features). Adding image features (DL, fusion) reduces bias at the cost of increased variance, mitigated by pretrained weights and regularization.

Clinical reliability. Temperature-scaled calibration ensures that the reported probability $\hat{q}_i$ (Equation 2) is a trustworthy estimate of malignancy risk, which is a prerequisite for clinical decision-support use.

7 Conclusion

We presented a systematic comparison of four approaches to binary skin cancer detection on the ISIC 2024 SLICE-3D dataset. Key takeaways:

1. **Patient-aware cross-validation is essential** — random splits inflate performance estimates by 5–15% in our experiments.
2. **pAUC @ TPR** ≥ 0.80 **is the correct primary metric** for clinical screening — standard AUC permits models that fail at high sensitivity.

3. **Focal Loss consistently outperforms weighted BCE** for this level of class imbalance.
4. **Temperature scaling materially improves calibration** (Expected Calibration Error reduced from [X] to [Y]), enabling trustworthy probability outputs for clinical decision support.
5. **Hybrid fusion model assessment** — whether the hybrid model outperforms the image-only baseline will be determined following full training, indicating whether metadata adds discriminative signal beyond image features.

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